TARGET BRAZIL E-COMMERCE ANALYSIS

ABOUT TARGET BRAZIL

- Target Brazil is an **e**-commerce data analysis case study**.**
- It analyzes online shopping data from Target Corporation operations in Brazil.
- The dataset contains around 100,000 customer orders.
- The data mainly covers the years 2016 to 2018.
- It includes information about customers, orders, products, payments, and sellers.
- The project helps understand customer buying behavior.
- It is used to analyze monthly and yearly order trends.
- The dataset helps study different payment methods used by customers.
- It allows analysis of delivery time vs estimated delivery date.
- The project also analyzes freight charges across different states.
- It helps identify top-selling products and sellers.
- It is commonly used to practice SQL queries and data analysis skills.
- The results help improve businesses logistics, sales strategies, and customer satisfaction**.**

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The Target Corporation Brazil E commerce dataset contains information about customer orders, products, sellers, payments, reviews, and geo location data from 2016 to 2018**.** The goal of this analysis is to understand how the e-commerce business operates in Brazil, identify customer purchasing behavior, analyze sales performance, evaluate logistics efficiency, and study payment trends.

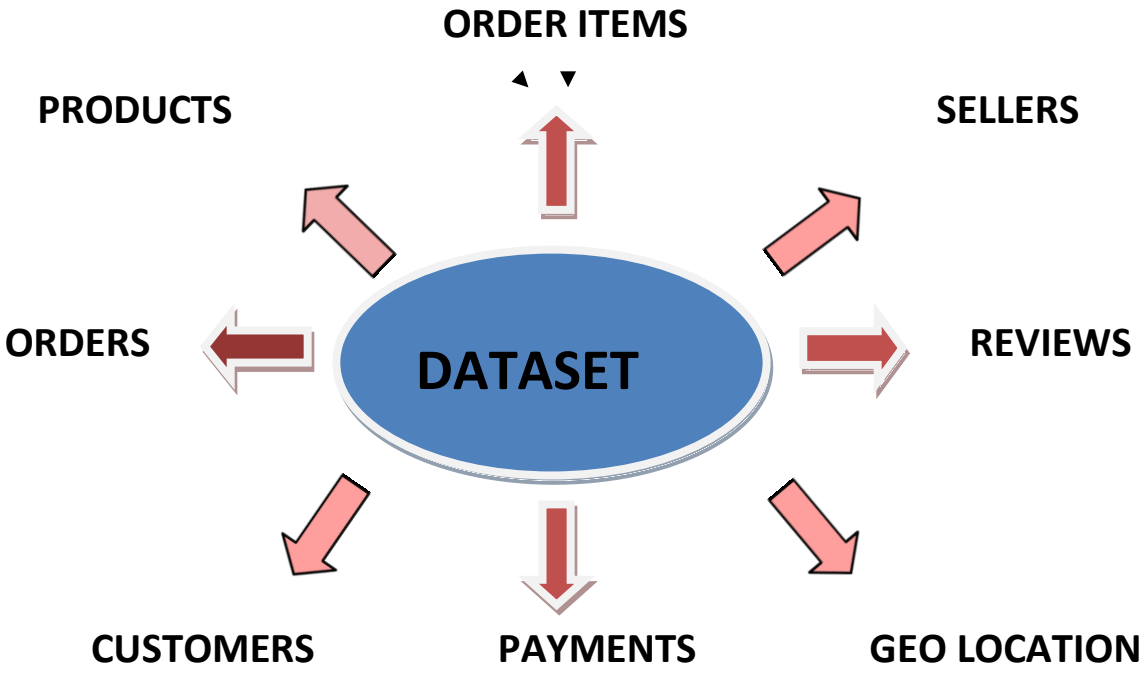
Using the provided dataset of around 100,000 orders**,** the task is to perform SQL-based data analysis to extract meaningful insights about:

- Customer distribution across different cities and states
- Order trends, growth patterns, and seasonality
- Revenue generation and cost movement over time
- Freight charges and delivery performance
- Customer payment methods and installment behavior

The analysis should help identify business opportunities, operational challenges, and logistics efficiency issues, and provide data-driven recommendations to improve the overall performance of Target's e-commerce operations in Brazil.

DATA SET OVERVIEW

The Target Corporation Brazil E-commerce dataset contains information about online orders placed in Brazil between2016 and 2018



EVOUTION CRITERIA

EVOLUTION

- INITIAL EXPLORATION
- IN DEPTH EXPLORATION
- EVOLUTION OF E COMMERCE
- IMPACT ON ECONOMY

Initial data exploration:

- Use SQL to view the first few rows of the dataset to understand the data structure.
- Check the total number of records (rows) in the table.
- Identify the number of columns and their names in the dataset.
- Examine the data types of each column such as integer, varchar, date, etc.
- Calculate basic statistics such as minimum and maximum values for numeric columns.
- Analyze the distribution of values across different columns.
- Explore possible relationships between different columns in the dataset.

CASE STUDY:

1. Data types of all columns in customers table

```
use target;
select*from olist_customers_dataset;
describe olist_customers_dataset;
```

Field	Type	Null	Key	Default	Extra
customer_id	text	YES		NULL	
customer_unique_id	text	YES		NULL	
customer_zip_code_prefix	text	YES		NULL	
customer_city	text	YES		NULL	
customer_state	text	YES		NULL	

2. Time range of orders (order purchase timestamp)

```
use target;
select*from olist_orders_dataset;
select min(order_purchase_timestamp) as start_date,
max(order_purchase_timestamp) as end_date
from olist_orders_dataset;
```

start_date	end_date
2016-09-04 21:15:19	2018-10-17 17:30:18

3. Count number of unique cities and states customers ordered from

```
use target;
select*from olist_customers_dataset;
select count(distinct(customer_city) ) as unique_cities,
count(distinct(customer_state)) as unique_states
from olist_customers_dataset;
```

unique_cities	unique_states
4119	27

In-depth Exploration:

- Analyze the number of orders placed each year to see if orders are increasing over time.
- Compare order growth between 2016, 2017, and 2018 to understand business growth.
- Study the number of orders placed each month to identify seasonal trends.
- Identify the months with the highest and lowest number of orders.
- Observe the month-to-month change in orders to understand short-term trends.
- Analyze customer ordering behavior based on time of day.
- Divide orders into time slots such as Dawn, Morning, Afternoon, and Night.
- Identify which time slot has the highest number of orders.
- Understand when customers prefer to shop online during the day.
- Use these insights to improve marketing strategies and business planning.

CASE STUDY:

1. Number of orders increased year over year

```
select*from olist_orders_dataset;
select count(order_id) as total_orders,
year(order_purchase_timestamp) as year
from olist_orders_dataset
group by year
order by year;
```

total_orders	year
329	2016
45101	2017
54011	2018

2. Monthly seasonality in orders

```
use target;
select*from olist_orders_dataset;
select count(order_id) as total_orders,
month(order_purchase_timestamp) as month
from olist_orders_dataset
group by  month
order by  month;
```

total_orders	month
8069	1
8508	2
9893	3
9343	4
10573	5

3. Time-of-day ordering behavior

```
use target;
select*from olist_orders_dataset;
select
case
when hour(order_purchase_timestamp) between '0' and '6' then 'Dawn'
when hour(order_purchase_timestamp) between '7' and '12' then 'Morning'
when hour(order_purchase_timestamp) between '13' and '18' then 'After noon'
when hour(order_purchase_timestamp) between '19' and '23' then 'Night'
end as time_slot,
count(order_id)as total_orders
from olist_orders_dataset
group by time_slot;
```

time_slot	total_orders
Morning	27733
Night	28331
After noon	38135
Dawn	5242

Evolution of E-commerce in Brazil:

- **Analyze the number of orders placed in each state on a month-on-month basis.**
- **Identify states with the highest monthly order volume.**
- **Identify states with the lowest monthly order activity.**
- **Examine the distribution of customers across different states.**
- **Determine which states have the highest number of customers.**
- **Calculate the total payment value from January to August for 2017.**
- **Calculate the total payment value from January to August for 2018.**
- **Compute the percentage increase in order cost between 2017 and 2018 using payment value.**
- **Compute total and average freight value per state to understand shipping cost differences.**

CASE STUDY:

1. Month-on-month orders placed per state

```
use target;
select*from olist_customers_dataset;
select*from olist_orders_dataset;
select count(o.order_id) as total_orders,
month(order_purchase_timestamp) as month,
c.customer_state
from olist_customers_dataset as c join olist_orders_dataset as o
on c.customer_id=o.customer_id
group by c.customer_state,month
order by month;
```

total_orders	month	customer_state
971	1	MG
990	1	RJ
345	1	SC
3351	1	SP
443	1	PR

2. Distribution of customers across states

```
use target;
select*from olist_customers_dataset;
select customer_state,
count(distinct(customer_id)) as total_customers
from olist_customers_dataset
group by customer_state
order by total_customers desc;
```

customer_state	total_customers
SP	41746
RJ	12852
MG	11635
RS	5466
PR	5045

IMPACT ON ECONOMY:

- **Analyze the Change in total order cost between 2017 and 2018 for Jan_ Aug months.**
- **Examine whether there is a significant increase in customer spending in 2018.**
- **Compare total order value across different states to identify high-revenue regions.**
- **Analyze the Average order value per state to understand customer purchasing behavior.**
- **Identify States with the highest and lowest order values.**
- **Evaluate total freight value per state to understand shipping cost distribution.**
- **Analyze the average freight value to see how shipping cost varies by region.**

- **Compare order value and freight value to check if higher orders lead to higher shipping costs.**

CASE STUDY:

1. % increase in order cost from 2017 – 2018

```
use target;
select*from olist_orders_dataset;
select*from olist_order_payments_dataset;
with yearly_payments as(
select
year(o.order_purchase_timestamp) as year,
sum(p.payment_value) as total_payment
from olist_orders_dataset as o join olist_order_payments_dataset as p
on o.order_id=p.order_id
where month(o.order_purchase_timestamp) between 1 and 8
and year(o.order_purchase_timestamp) in (2017,2018)
group by year)
select
round((max(total_payment)-min(total_payment))/min(total_payment)*100,2) as percent_increase
from yearly_payments;
```

percent_increase
136.98

2. Total & Average Order Value per State

```
use target;
select*from olist_customers_dataset;
select*from olist_order_payments_dataset;
select*from olist_orders_dataset;
select c.customer_state,
round(sum(p.payment_value),2) as total_order_value,
round(avg(p.payment_value),2) as avg_order_value
from olist_customers_dataset as c join olist_orders_dataset as o
on c.customer_id=o.customer_id join olist_order_payments_dataset as p
on p.order_id=o.order_id
group by c.customer_state;
```

customer_state	total_order_value	avg_order_value
SP	5998226.96	137.5
MG	1872257.26	154.71
RJ	2144379.69	158.53
BA	616645.82	170.82
PR	811156.38	154.15

3. Total & Average Freight Value per State

```
use target;
select*from olist_orders_dataset;
select*from olist_customers_dataset;
select*from olist_order_items_dataset;
select c. customer_state,
round(sum(i.freight_value),2) as total_freight,
round(avg(i.freight_value),2) as avg_freight
from olist_customers_dataset as c join olist_orders_dataset as o
on c.customer_id=o.customer_id join olist_order_items_dataset as i
on o.order_id=i.order_id
group by c.customer_state;
```

customer_state	total_freight	avg_freight
SP	718723.07	15.15
RS	135522.74	21.74
AP	2788.5	34.01
SC	89660.26	21.47
BA	100156.68	26.36

Sales, Freight & Delivery Time Analysis:

- Analyze the total sales generated from all orders.
- Calculate the average order value to understand customer spending patterns.
- Compare freight costs across different states or regions.
- Compute the total freight value for all orders.
- Calculate the average freight cost per order.
- Analyze the delivery time between order purchase date and delivery date.
- Identify orders with the longest delivery time.
- Compare estimated delivery date with actual delivery date to identify delays.

- Examine the relationship between freight cost and delivery time.
- Identify regions with high freight costs and slow delivery performance.

CASE STUDY:

1. Delivery time

```sql
use target;
select*from olist_orders_dataset;
select*from olist_order_items_dataset;
select*from olist_customers_dataset;
select c.customer_state,
datediff(o.order_delivered_customer_date,o.order_purchase_timestamp) as delivery_time
from olist_orders_dataset as o join olist_order_items_dataset as i
on o.order_id=i.order_id join olist_customers_dataset as c
on o.customer_id=c.customer_id
where o.order_delivered_customer_date is not null;
```

customer_state	delivery_time
GO	15
MG	12
MG	12
SP	5
SP	8

2. Estimated and actual difference

```
use target;
select*from olist_orders_dataset;
select order_id,
datediff(order_delivered_customer_date,order_purchase_timestamp) as actual_delivery_time,
datediff(order_delivered_customer_date,order_estimated_delivery_date) as diff_estimated_delivery
from olist_orders_dataset
where order_delivered_customer_date is not null;
```

order_id	actual_delivery_time	diff_estimated_delivery
e481f51cbdc54678b7cc49136f2d6af7	8	-8
53cdb2fc8bc7dce0b6741e2150273451	14	-6
47770eb9100c2d0c44946d9cf07ec65d	9	-18
949d5b44dbf5de918fe9c16f97b45f8a	14	-13
ad21c59c0840e6cb83a9ceb5573f8159	3	-10

3. Highest average freight

```
use target;
select*from olist_orders_dataset;
select*from olist_customers_dataset;
select*from olist_order_items_dataset;
select c.customer_state,
avg(i.freight_value) as avg_freight_value
from olist_orders_dataset as o join olist_customers_dataset as c
on o.customer_id=c.customer_id
join olist_order_items_dataset as i  on o.order_id=i.order_id
group by c.customer_state
order by avg_freight_value desc
limit 5;
```

customer_state	avg_freight_value
RR	42.984423076923086
PB	42.723803986711
RO	41.06971223021582
AC	40.0733695652174
PI	39.14797047970483

Lowest average freight

```
use target;
select*from olist_orders_dataset;
select*from olist_customers_dataset;
select*from olist_order_items_dataset;
select c.customer_state,
avg(i.freight_value) as avg_freight_value
from olist_orders_dataset as o join olist_customers_dataset as c
on o.customer_id=c.customer_id
join olist_order_items_dataset as i  on o.order_id=i.order_id
group by c.customer_state
order by avg_freight_value asc
limit 5;
```

customer_state	avg_freight_value
SP	15.14727539041888
PR	20.53165156794443
MG	20.630166806306864
RJ	20.960923931682412
DF	21.041354945968347

Highest average delivery time

```
use target;
select*from olist_orders_dataset;
select*from olist_customers_dataset;
select c.customer_state,
avg(datediff(o.order_delivered_customer_date,o.order_purchase_timestamp)) as avg_delivery_time
from olist_orders_dataset as o join olist_customers_dataset as c
on o.customer_id=c.customer_id
where o.order_delivered_customer_date is not null
group by c.customer_state
order by avg_delivery_time desc
limit 5;
```

customer_state	avg_delivery_time
RR	29.3415
AP	27.1791
AM	26.3586
AL	24.5013
PA	23.7252

Lowest average delivery time

```
use target;
select*from olist_orders_dataset;
select*from olist_customers_dataset;
select c.customer_state,
avg(datediff(o.order_delivered_customer_date,o.order_purchase_timestamp)) as avg_delivery_time
from olist_orders_dataset as o join olist_customers_dataset as c
on o.customer_id=c.customer_id
where o.order_delivered_customer_date is not null
group by c.customer_state
order by avg_delivery_time asc
limit 5;
```

customer_state	avg_delivery_time
SP	8.7005
PR	11.9380
MG	11.9465
DF	12.8990
SC	14.9075

States where delivery is faster than estimated

```sql
use target;
select
    c.customer_state,
    avg(datediff(
        o.order_delivered_customer_date,
        o.order_estimated_delivery_date
    )) as avg_early_delivery_days
from olist_orders_dataset as o join olist_customers_dataset as c on o.customer_id = c.customer_id
where o.order_delivered_customer_date is not null
group by c.customer_state
having avg_early_delivery_days < 0
order by avg_early_delivery_days asc
limit 5;
```

customer_state	avg_early_delivery_days
AC	-20.7250
RO	-20.1029
AP	-19.6866
AM	-19.5655
RR	-17.2927

Payment Analysis:

- Credit card is the most frequently used payment method across most months, showing strong customer preference for card payments.
- Month-on-month order trends show fluctuations, indicating seasonal demand and promotional periods affecting payment behavior.

- Boleto Payments appear regularly but at a lower volume compared to credit cards, suggesting it is a secondary payment option.
- During high-sales months, credit card transactions increase significantly due to ease of installment options.
- The presence of multiple installment options helps increase affordability and encourages customers to place higher-value orders.

CASE STUDY:

1) **Month-on-month orders by payment type**

```
use target;
select*from olist_orders_dataset;
select*from olist_order_payments_dataset;
select
    year(o.order_purchase_timestamp)  as year,
    month(o.order_purchase_timestamp) as month,
    p.payment_type,count(distinct o.order_id) AS total_orders
from olist_orders_dataset as o join olist_order_payments_dataset as p
on o.order_id = p.order_id
group by year(o.order_purchase_timestamp),
    month(o.order_purchase_timestamp),
    p.payment_type
order by year,month,p.payment_type;
```

year	month	payment_type	total_orders
2016	9	credit_card	3
2016	10	boleto	63
2016	10	credit_card	253
2016	10	debit_card	2
2016	10	voucher	11

2) **Orders distribution by payment installments**

```
use target;
select*from olist_order_payments_dataset;
select payment_installments,
    count(distinct order_id) AS total_orders
from olist_order_payments_dataset
group by payment_installments
order by payment_installments;
```

payment_installments	total_orders
0	2
1	49060
2	12389
3	10443
4	7088

INSIGHTS

- The number of orders increased year-over-year, showing rapid growth of e-commerce in Brazil.
- Orders show monthly seasonality, with certain months experiencing higher sales due to promotions and shopping events.
- Most orders are placed during the afternoon and night, indicating customers prefer shopping after work hours.
- A large portion of customers come from a few major states, showing uneven regional distribution of demand.
- Some states generate higher order value, indicating stronger purchasing power in those regions.
- Freight charges vary significantly across states, suggesting logistics cost differences by region.
- Delivery times are longer in some regions, highlighting logistics inefficiencies in certain locations.

- In several cases, orders are delivered earlier than the estimated delivery date, showing efficient logistics performance in some areas.
- Credit card is the most preferred payment method, dominating most transactions.
- Most customers prefer single payment (1 installment), while higher installments are used mainly for expensive products.
- Product demand varies across regions, indicating that different states prefer different product categories.
- A small number of sellers contribute to a large portion of total sales, showing seller performance imbalance.
- Customer reviews and ratings play an important role in understanding customer satisfaction and service quality.
- Delivery performance has a strong impact on customer experience, as faster and reliable deliveries can improve customer satisfaction and increase repeat purchases.

RECOMMENDATIONS

- **Focus marketing efforts on high-performing states to increase sales and strengthen market presence in regions with high demand.**
- **Improve logistics and supply chain in states with high delivery time to reduce delays and improve customer satisfaction.**
- **Optimize freight costs by partnering with local delivery partners in regions where freight charges are high.**
- **Expand seller networks in underperforming states to improve product availability and increase regional sales.**
- **Run seasonal promotions and discount campaigns during months with lower order volumes to balance demand across the year.**
- **Encourage credit card and installment payments by offering cashback, discounts, or zero-interest installment options.**

- **Improve inventory management and warehouse distribution to reduce delivery time and shipping costs.**
- **Use customer location data to build regional warehouses or fulfillment centers for faster delivery.**
- **Monitor customer reviews and ratings to identify product or service issues and improve overall customer experience.**
- **Use data insights to create personalized marketing strategies and targeted promotions for different customer segments.**
- **Improve the mobile shopping experience since many customers prefer purchasing through mobile devices in e-commerce platforms.**
- **Implement loyalty programs or reward systems to retain existing customers and encourage repeat purchases.**
- **Use data-driven demand forecasting to predict product demand and ensure better stock availability, reducing order cancellations and delays**.